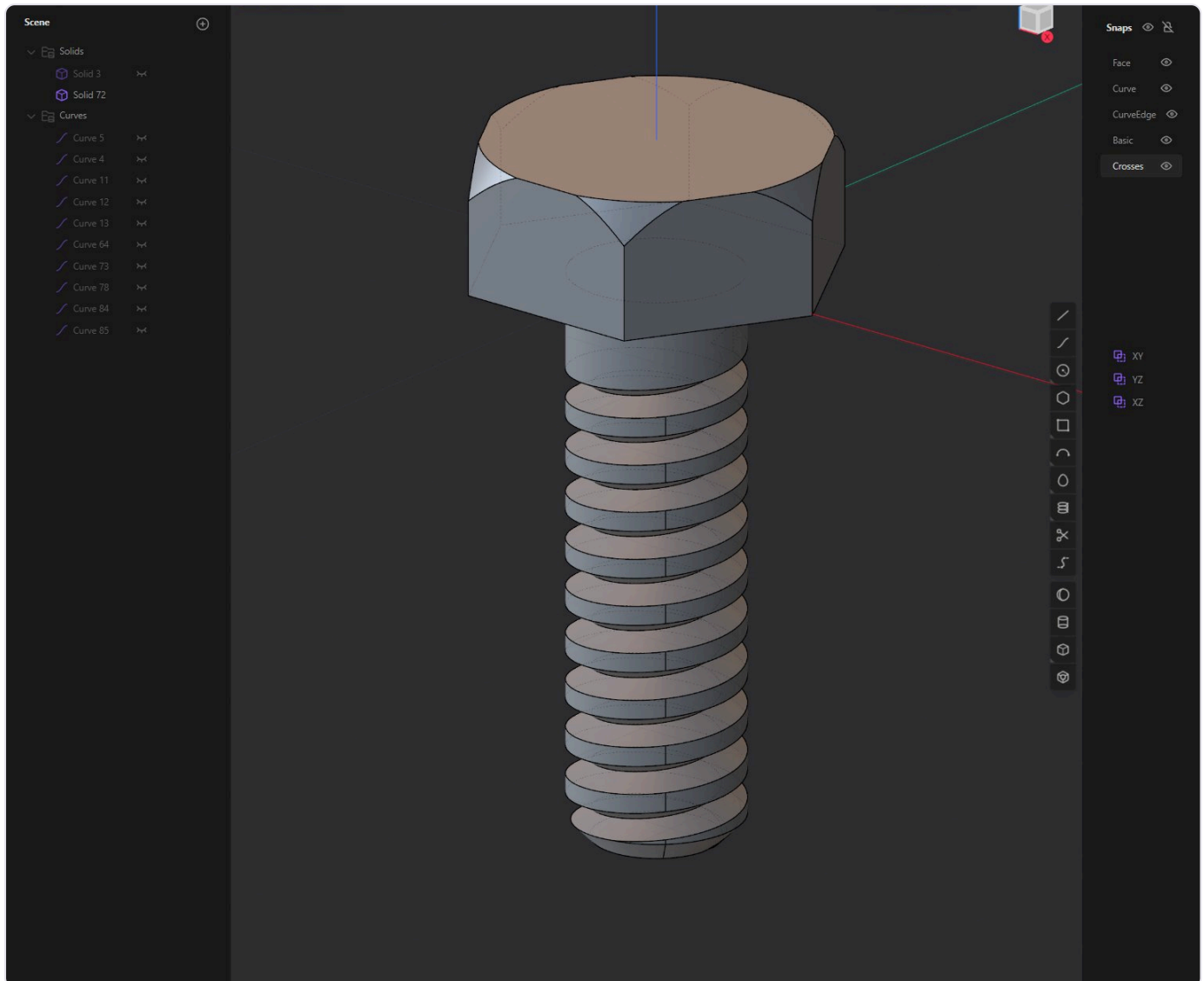


Modeling a Threaded Hex Screw

A step-by-step tutorial for Solidify, from a hexagon sketch to a cut thread



RESULT

Hex-head screw with a chamfered head, chamfered tip and a 60° thread

WORKFLOW

The usual FreeCAD/Fusion screw tutorial: extrude, intersect, sweep a profile along a helix, subtract

SIZE

Head 2 units across corners, shank radius 0.45 units, length 3 units, pitch 0.25 units

Tools used

Every tool below is in the right-hand toolbar or the bottom toolbar, and most have a keyboard shortcut. The shortcuts in a command only work while the mouse is over the viewport.

TOOL	WHERE / SHORTCUT	USED FOR
Regular polygon	Right toolbar (hexagon icon). Shift + wheel adds or removes sides.	Hexagon for the head
Center and radius circle	Right toolbar, Shift + W	Chamfer cone and shank profiles
Extrude	Bottom toolbar, E . While extruding: D distance, A angle, Q / W / E / R union/difference/intersect/new body.	Head, chamfer cone, shank
Boolean	Q , then Q union, W difference, E intersect	Rounding the head corners, cutting the thread
Fillet / chamfer	B . A negative distance makes a chamfer.	Tip chamfer
Hide / unhide	H hides the selection; the eye icon in the Scene list shows it again (Alt + H shows everything)	Drawing helper curves without snapping to the body
Construction plane	XY / YZ / XZ in the right-hand panel	Drawing the helix axis and the thread profile in the XZ plane
Spiral	Right toolbar (coil icon). While drawing: D length, R radius, A angle.	The helix the thread follows
Line	Right toolbar, Shift + A . Click the first point again to close the shape.	Triangular thread profile
Sweep (Evolution)	Bottom toolbar, Alt + E	Sweeping the profile along the helix

General keys: **Enter** or **OK** confirms a command, **Esc** cancels it, and **Ctrl** + **Z** undoes. Values are typed into the dialog at the bottom left. Lengths are in scene units. The extrude "Race" angle is in radians ($-0.52 \text{ rad} = -30^\circ$).

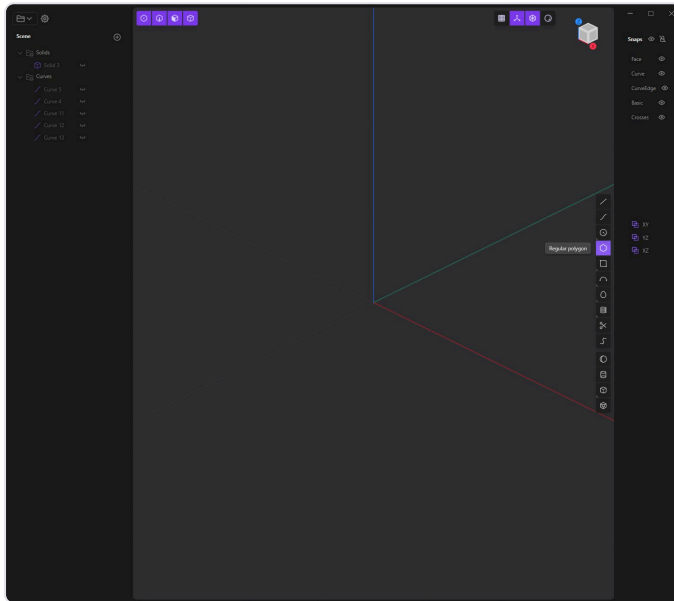
Overview

1. Draw a hexagon and extrude it into the head.
2. Round the head's corners: intersect it with a 30° cone.
3. Extrude the shank from the underside of the head and join it.
4. Chamfer the tip.
5. Draw a helix around the shank with the Spiral tool.
6. Draw a 60° triangle at the start of the helix.
7. Sweep the triangle along the helix into a coil.
8. Subtract the coil from the screw to cut the thread.

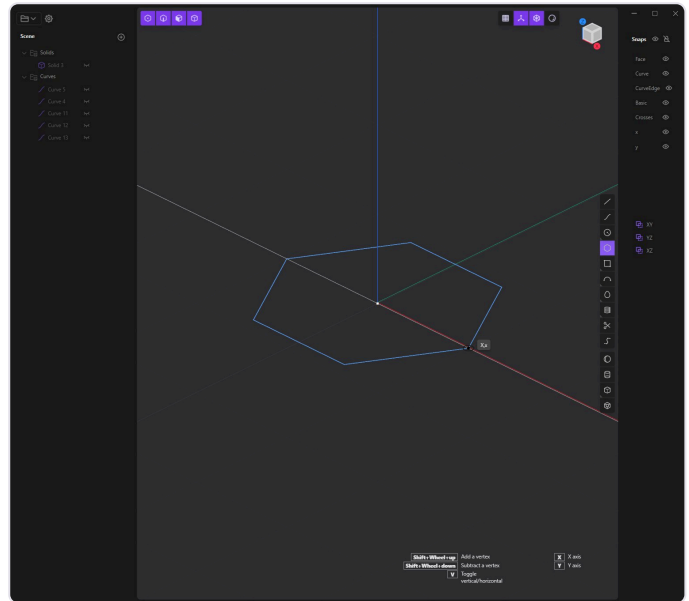
1 Draw the hexagon

Regular polygon

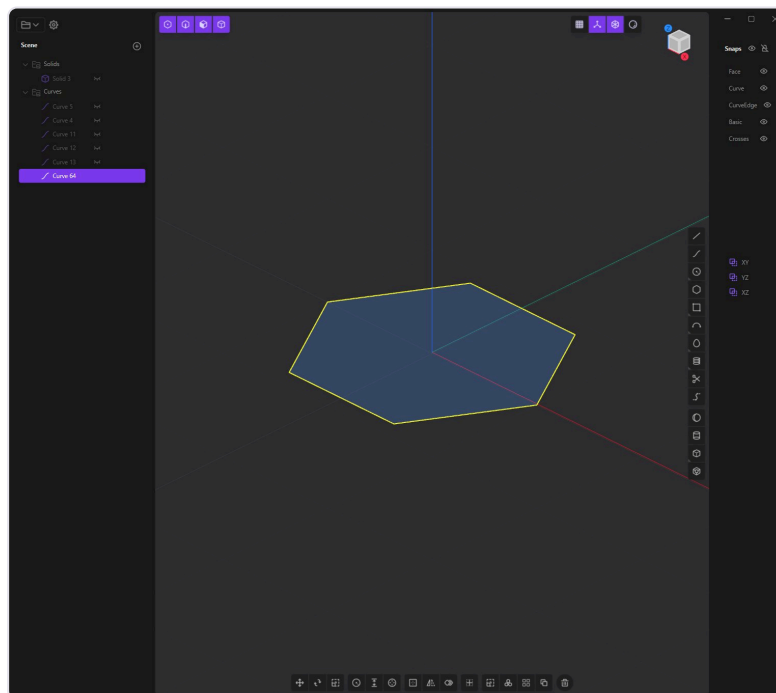
1. In the right-hand toolbar, click **Regular polygon** (the hexagon icon).
2. Click the origin to place the center.
3. Move the mouse out along the red X axis. The polygon starts with 5 sides: hold **Shift** and scroll the wheel up once to get 6.
4. Click 1 unit from the center. A closed curve becomes a **region**, which can be extruded.



The Regular polygon tool in the right-hand toolbar.



Drawing it: the hints at the bottom right show **Shift** + wheel for adding or removing vertices.

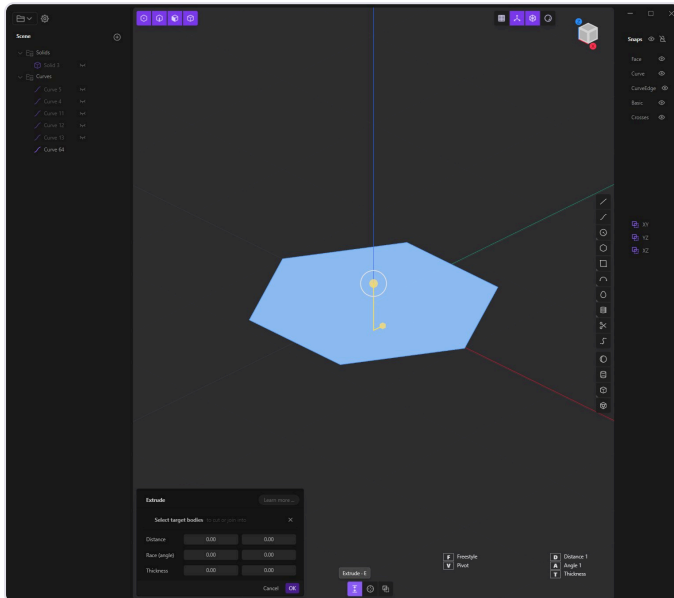


The finished hexagon region (1 unit from center to corner).

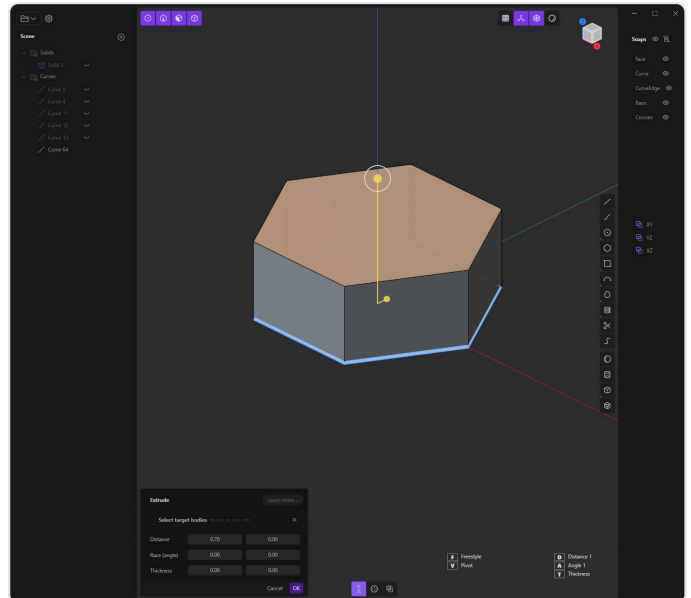
2 Extrude the head

Extrude · E

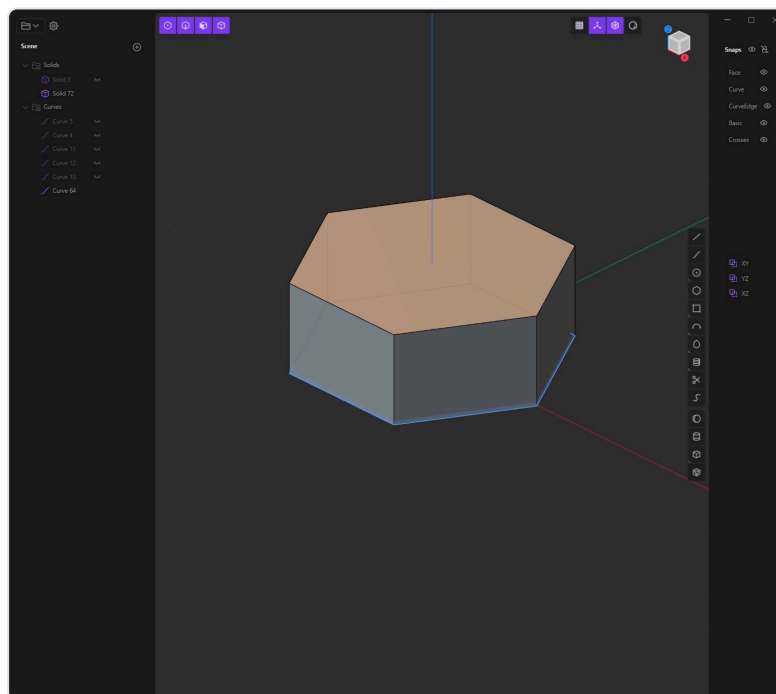
1. Click inside the hexagon to select the region.
2. Press **E**, or click **Extrude** in the bottom toolbar.
3. In the dialog, set the first **Distance** to **0.70** (upwards), then click **OK**.



Extrude (**E**) on the selected region.



Distance 0.70: preview with the drag arrow.



The hexagonal head.

3 Round the head's corners

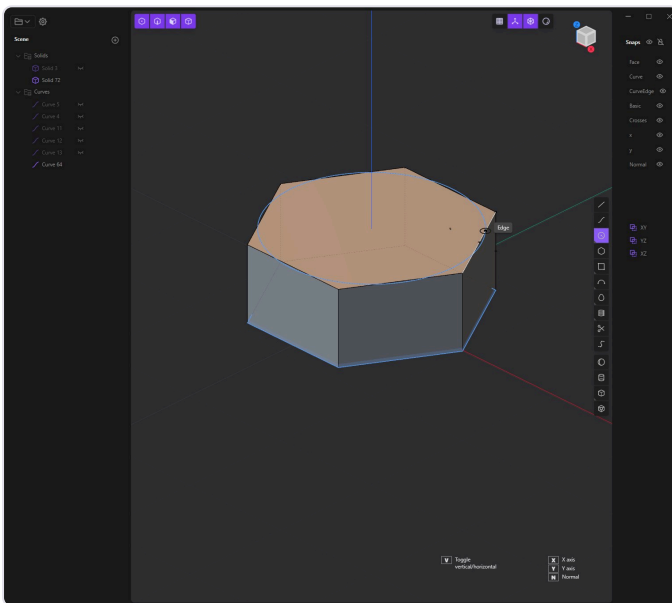
Center and radius circle · Shift+W

Extrude with angle

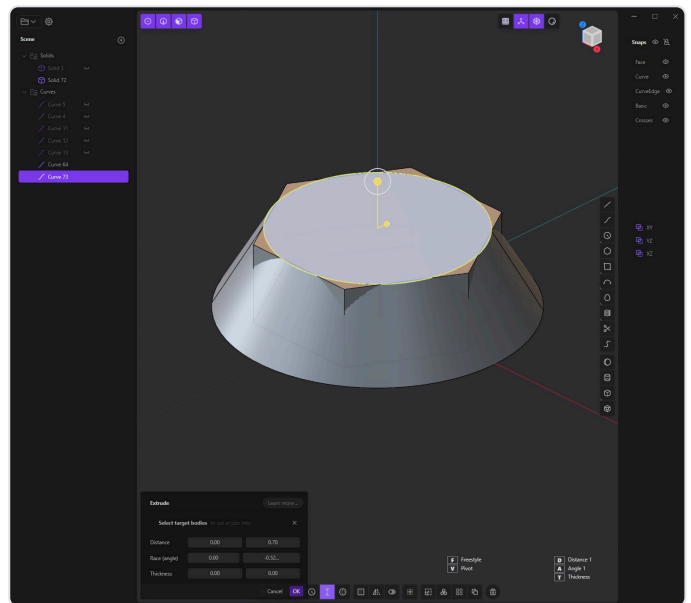
Boolean · Q → E (Intersect)

Real bolt heads have their corners trimmed by a 30° cone, which leaves curved arcs on each flat. We build that cone and keep only the part of the head inside it.

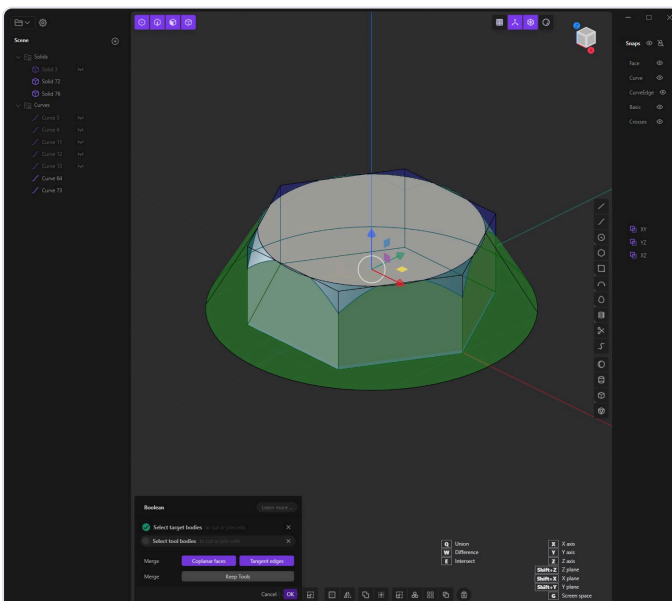
1. Press **Shift** + **W**. Click the center of the top face, then click about 0.89 units out, just beyond the middle of the flats.
2. Select only this circle region and press **E**. Set **Distance** to **0.00** in the first box and **0.70** in the second, so it goes down through the head. Set the second **Race (angle)** box to **-0.52** rad (-30°) so the cone widens downwards. Click **OK**, which creates a separate cone body.
3. Select the head, then **Shift**-click the cone. Press **Q** (Boolean) and then **E** for **Intersect**; the preview turns green. Click **OK**.



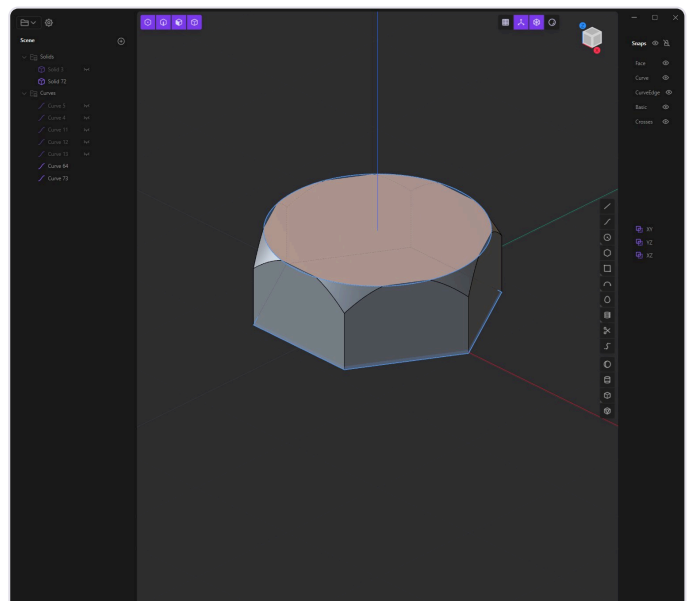
The circle on the top face snaps to the face.



Extruded down 0.70 with a -0.52 rad angle: a cone.



Boolean, Intersect (**E**): the green preview is what remains.



The head with its corners rounded off.

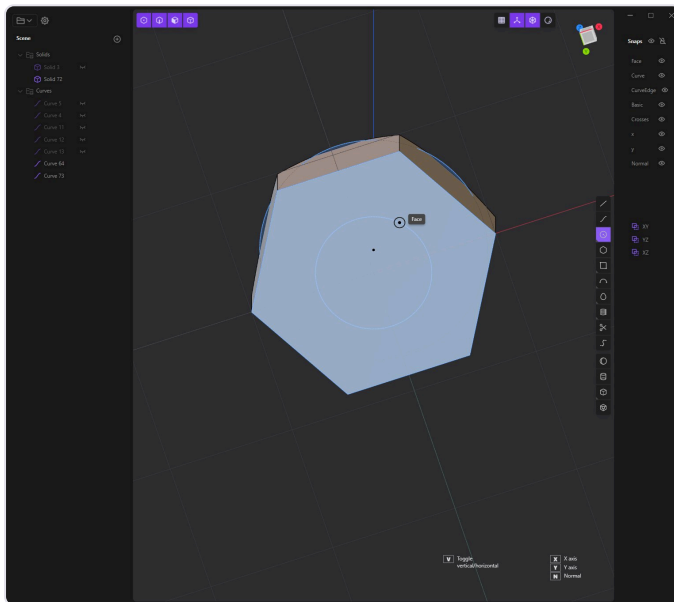
4 Extrude the shank

Circle · Shift+W

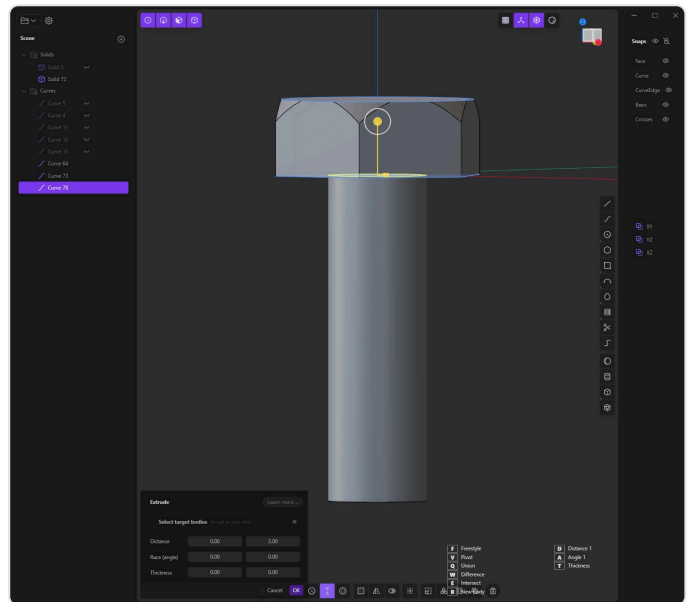
Extrude · E → Q (Union)

1. Orbit the view (middle-mouse drag) until you look at the underside of the head.
2. Press **Shift** + **W**, click the center of the bottom face and draw a circle of radius **0.45**.
3. Select the circle region and **Shift**-click the head, then press **E**.
4. Drag the arrow a little (or press **D**, move and click). This switches on the union, difference, intersect and new-body keys.
5. Set **Distance** to **0.00** in the first box and **3.00** in the second, then press **Q** for **Union** and click **OK**.

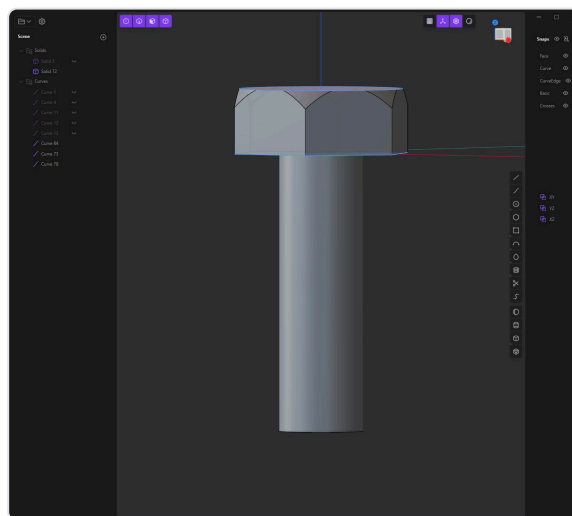
When an extrusion touches a body, it starts out as a cut (red preview). Press **Q** to join it to the body instead.



Circle on the underside (the face snap is shown).



Extruded 3.00 downwards; the union, difference and intersect keys are listed bottom right.

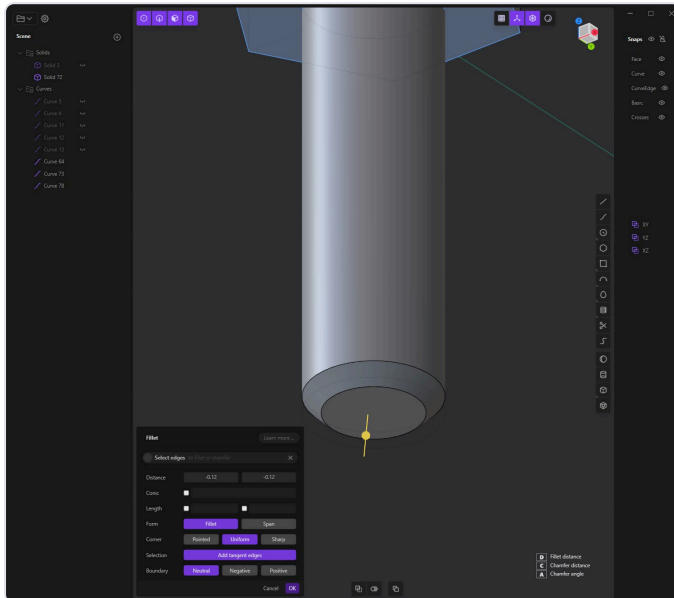


Head and shank are now one body.

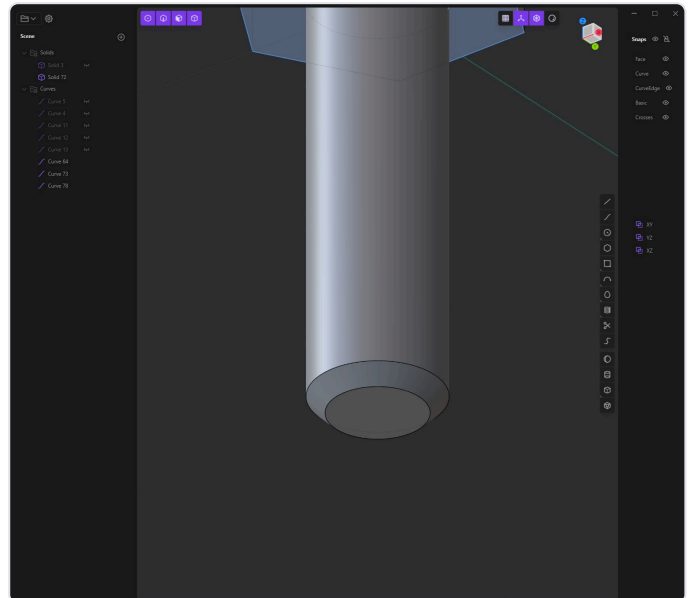
5 Chamfer the tip

Fillet / chamfer - B

1. Select the circular edge at the bottom of the shank (edge selection mode: **2**).
2. Press **B**. Type **-0.12** into **both** Distance boxes; a negative distance makes a chamfer instead of a fillet.
3. Click **OK**.



Fillet dialog with -0.12 / -0.12: a chamfer.



The chamfered tip.

6 Draw the helix

Hide · H

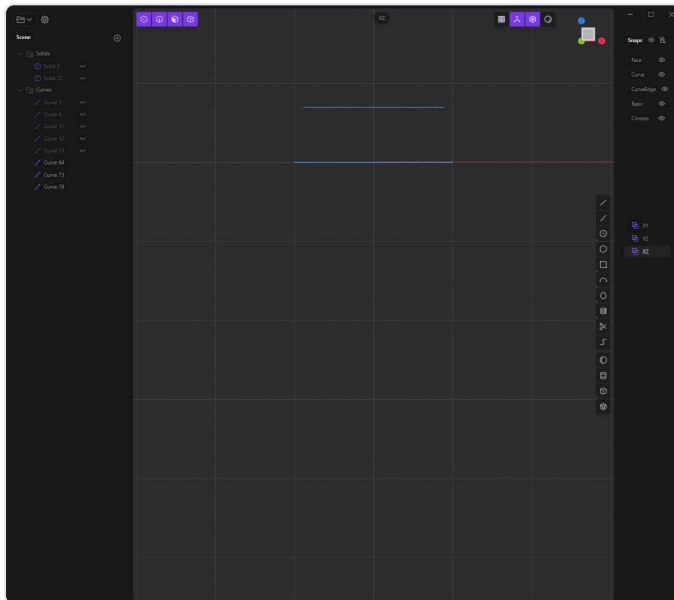
XZ construction plane

Spiral

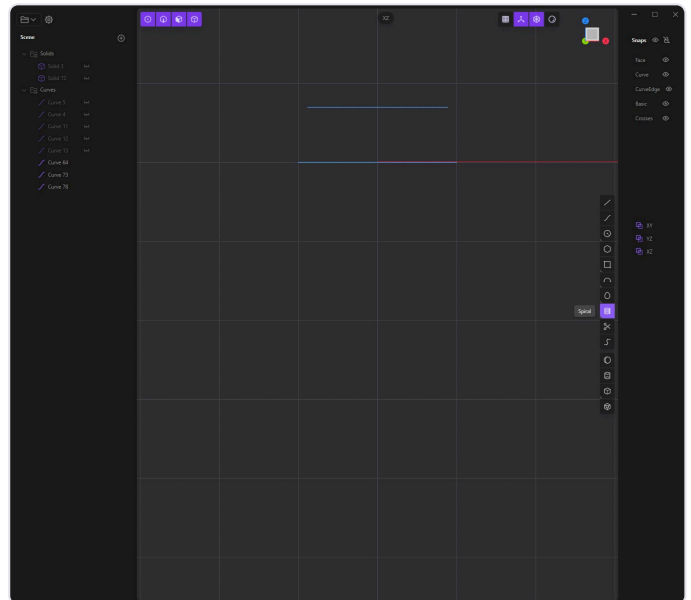
The thread follows a helix around the shank. Draw it in a front view on the XZ plane, so the clicks land exactly on the shank's axis.

1. Select the screw and press **H** to hide it, so your clicks don't snap to its faces.
2. Click **XZ** in the right-hand panel and look from the front.
3. Click **Spiral** in the right-hand toolbar.
4. Click the **axis start** just below the tip at (0, -3.25), then the **axis end** straight above it at (0, -0.45); the Z-axis snap keeps it vertical. Finally click the **radius** point to the right (+X).
5. In the dialog set **Step** (the number of turns) to **11.20** and **Radius** to **0.4536**, the shank radius. That gives a pitch of $2.8 / 11.2 = 0.25$. Press **Enter**.

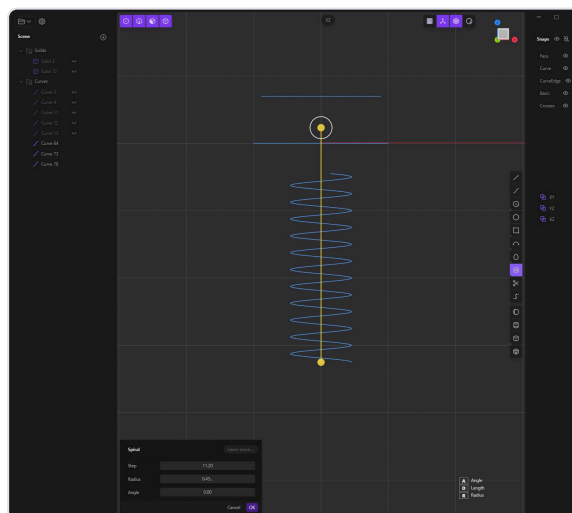
The helix starts below the tip so the thread runs out cleanly at the end. It stops 0.45 below the head, leaving a plain section under the head.



Screw hidden; the XZ plane is active (highlighted at right).



The Spiral tool.



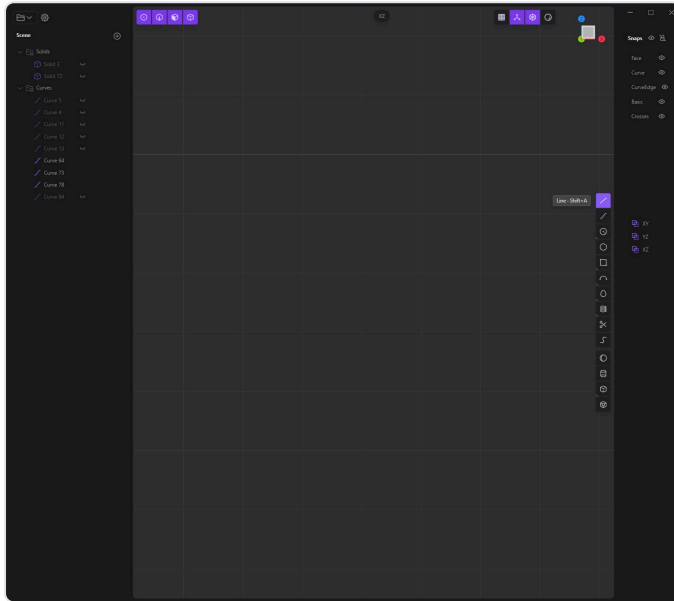
The helix: Step 11.20 turns, Radius 0.45. Use the handles or **D** / **R** / **A** to adjust length, radius and taper angle.

7 Draw the thread profile

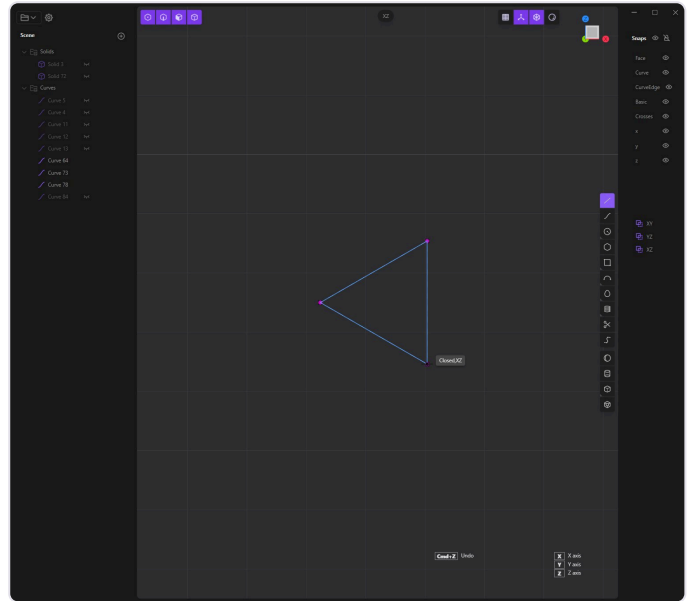
Line · Shift+A

The thread profile is a 60° triangle in the XZ plane at the helix's start point. Its base sits 0.05 outside the shank and its tip points 0.13 into it.

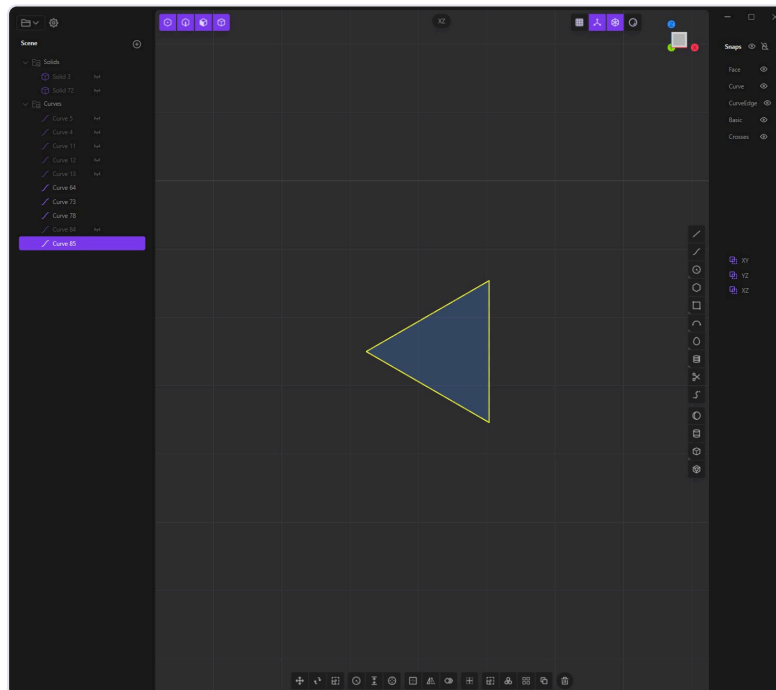
1. Select the helix and press **H**, so the line clicks don't snap onto it. Zoom in on the bottom of the axis.
2. Press **Shift** + **A** (Line) and click **(0.50, -3.35)**, **(0.50, -3.15)** and **(0.32, -3.25)**.
3. Click the first point again: the **Closed** snap closes the triangle, and it becomes a region.



The Line tool (**Shift** + **A**).



Closing the triangle on its first point ("Closed").



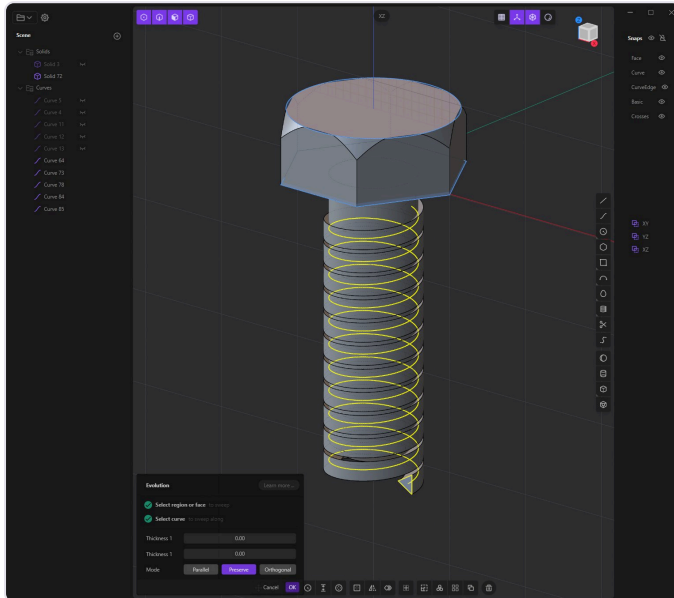
The closed thread profile.

8 Sweep the profile along the helix

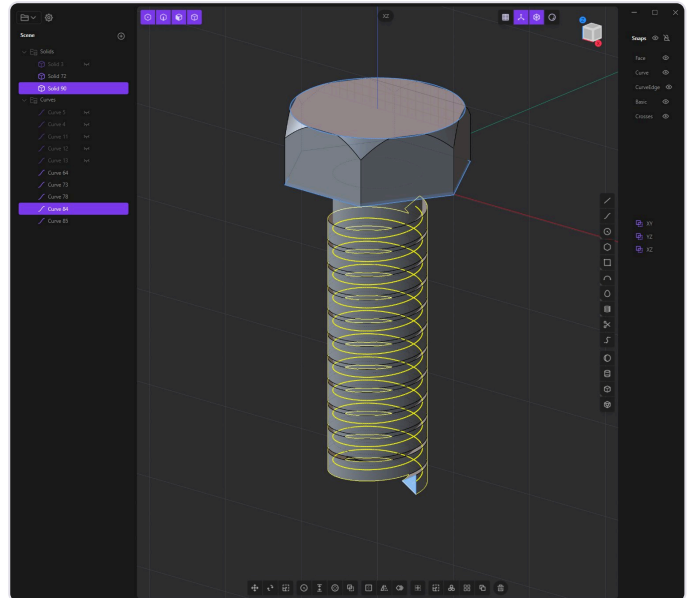
Unhide (eye icon)

Sweep · Alt+E

1. Show the screw and the helix again with their eye icons in the Scene list.
2. Select the triangle region, then **Shift**-click the helix.
3. Press **Alt** + **E** (Sweep). Keep **Mode: Preserve**; the profile then turns with the helix and stays in the axial plane. Press **Enter**.



Sweep dialog: region and curve selected, Mode Preserve.

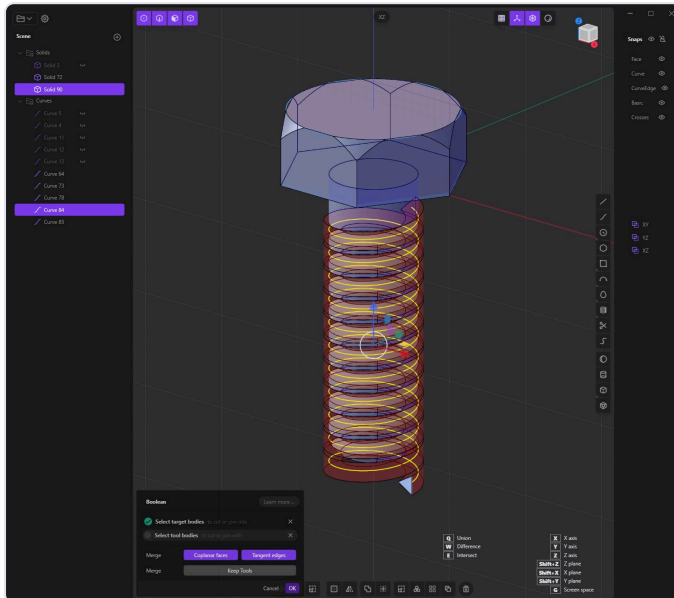


The swept coil around the shank.

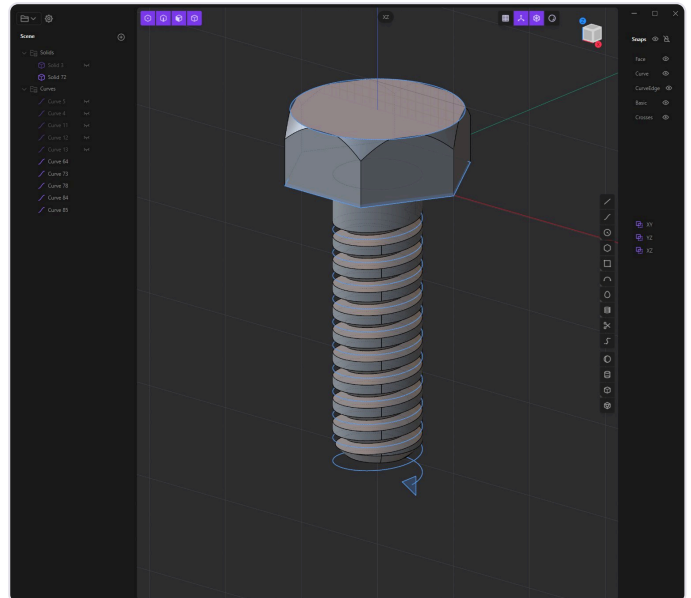
9 Cut the thread

Boolean · Q → W (Difference)

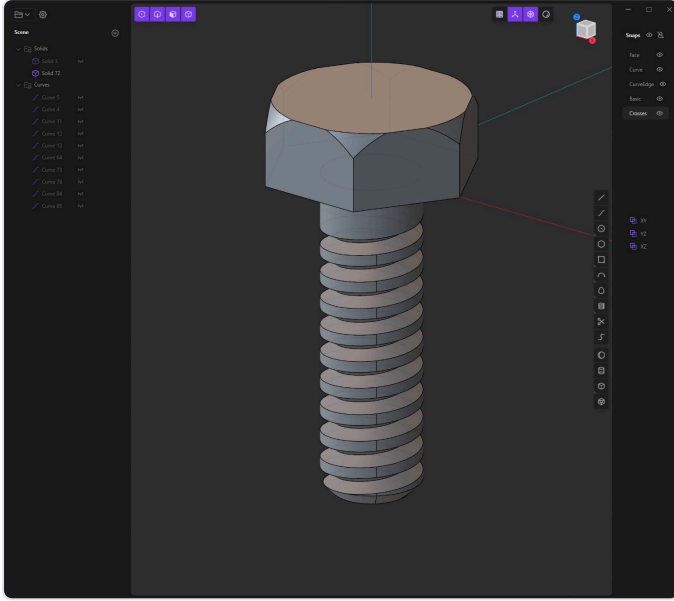
1. Select the screw, then **Shift**-click the coil.
2. Press **Q** (Boolean), then **W** for **Difference**; the coil turns red. Click **OK**.
3. Hide the sketch curves (select them in the Scene list and press **H**).



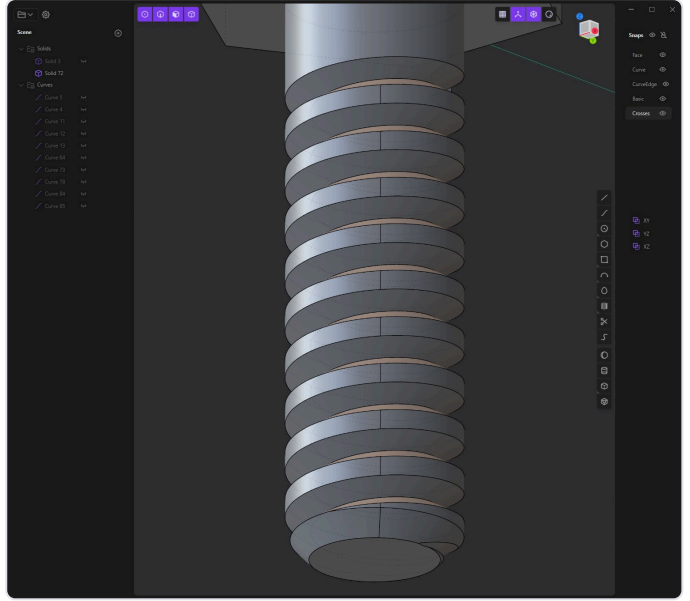
Boolean, Difference: the red coil is removed from the screw.



The thread is cut.



The finished hex screw.



Close-up: even thread, run-out below the head, chamfered tip.

Variations

- **Finer or coarser thread:** change the Spiral's Step (number of turns) and scale the triangle to match the pitch.
- **Tapered (wood-screw) thread:** give the Spiral an **Angle**; its radius then changes along the axis.
- **Socket head:** draw a hexagon on the top face and extrude it down as a cut (difference) instead of rounding the corners.